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Statistical phase space theory of polyatomic systems: Rigorous energy and angular momentum conservation in reactions involving symmetric polyatomic species

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Classical formulas for the sum and density of rotational and rotational-orbital states pairs of polyatomic molecules are derived using a geometrical approach that rigorously conserves energy and angular momentum. The pair combinations considered include a spherical top in combination with either a linear, spherical, or symmetric top molecule. With the formulas presented here it is possible to perform exact classical statistical phase space theory calculations for bimolecular reactions involving polyatomic molecules. In suitable limits our results reduce to well-known classical expressions for rotational state densities and sums or to equations previously derived by Klotz using an approximate phase space approach. Calculations are given that demonstrate the distortions produced in the system phase space by molecular oblateness and prolateness are usually small.

I. INTRODUCTION

In recent years some very exciting experimental results have been obtained regarding the detailed mechanism of chemical reactions.¹ The central question of energy and momentum disposal in collision processes is not only being asked in earnest, but in some cases it appears that answers are beginning to emerge. Most collision systems studied to date have been relatively simple (i.e., 3–5 atoms), but recently more work is being performed and reported on collisions involving polyatomic molecules. The vast majority of the definitive work on bimolecular collisions in neutral systems has been performed on beam machines,¹ although chemical activation studies have made substantial contributions to our understanding of unimolecular reactions.² Recent infrared chemiluminescence data has also shown promise as a mechanistic probe.³

Ionic systems show particular promise for testing and expanding our kinetic concepts. The principal advantages of these systems are that energy is often a convenient variable and ions are generally easy to detect. For example, rates of unimolecular fragmentations of ions have been reported for internal energy ranges of several eV above threshold, using both charge transfer spectroscopy^{4a} and photoelectron-photoion coincidence spectroscopy.^{4b} Photoionization techniques have been used to study the dependence of bimolecular reaction cross sections on internal energies⁵ and crossed beam techniques to measure product kinetic energy distributions of bimolecular reactions.^{6a} Advances in ion kinetic energy mass spectrometry have recently allowed measurement of kinetic energy distributions of unimolecular fragmentations.⁶

The data obtained by experimental studies of polyatomic systems requires an increased theoretical understanding of the detailed process involved in a collision between two polyatomic molecules, or a unimolecular decay event of a polyatomic species. One approach to the problem is to perform trajectory studies on a suitable potential surface and extract reaction transition probabilities from the trajectory results.⁷ This approach is very useful for small systems that do not form long lived complexes. At present, however,

such an approach is unmanageable for systems with much more than three atoms or for deep wellled systems forming long lived complexes that “trap” the trajectory.

An alternative approach is to appeal to the statistical hypothesis, and then test the validity of the hypothesis for a given system by comparison with experiment. The essence of the statistical hypothesis may be summarized in terms of the following three assumptions. First, assume that reaction proceeds through resonance states in which there is strong coupling of all degrees of freedom. This process is often commonly referred to as complex formation. Second, assume that in the passage of the reacting system from complex to products there is a critical configuration, called the transition state or activated complex, which governs the rate of product formation. Third, assume that the distribution of complex lifetimes is random. This assumption allows us to take the reaction probabilities, from complex to products, as proportional to the number of accessible states in the region of the transition state. It is the statistical approach that we will be concerned with in the present paper.

The origins of the statistical approach lie in the development of activated complex or transition state theory,⁸ and of the related RRKM theory of unimolecular decay.⁹ In the original formulations of these theories it was assumed that reaction occurred on the constant energy hypersurface of the system phase space, but not necessarily on the constant total angular momentum hypersurface. However the constraints placed on the transition state by angular momentum conservation may at times be severe, thus it is desirable to develop transition state-RRKM theory in a form which rigorously conserves both energy and angular momentum.

Approximate methods of angular momentum conservation for the RRKM theory of thermal unimolecular reactions are well known.¹⁰ The first successful attempt to apply rigorous angular momentum conservation in a statistical theory of bimolecular reactions was the “phase space” theory of Light, Pechukas and co-work-

ers¹¹ and Nikitin.¹² Although this theory was developed from the perspective of the reaction products it has been shown^{13,14} that, in the loose transition state theory and phase space theory are formally equivalent. The loose transition state limit assumes that the reaction potential surface is of negligible importance in determining the course of the reaction.

Phase space theory was first developed in quantum and classical form for systems composed of a diatomic molecule paired with an atom. The quantum and classical forms of the theory have been extended to systems composed of an atom and prolate or oblate top^{12,15} and to two diatomics.^{12,15a,16} To date angular momentum conservations in systems composed of two polyatomic species has been treated only in approximate forms. Klotz¹⁷ has developed a classical phase space approach for the system composed of two spherical species, under the condition that the system total angular momentum $\mathcal{J}$ is small. This approximation has restricted the utility of his equations to the domain of unimolecular kinetics. A transition state formalism, designed to predict product translational energy distributions for the limiting case in which $\mathcal{J}$ passes to product orbital angular momentum $\mathbf{L}$, has been developed by Safron *et al.*¹⁸ Marcus^{11a,19} has extended transition state theory to the limiting cases in which $\mathcal{J}$ passes entirely either to $\mathbf{L}$ or to product rotational angular momentum $\mathbf{J}_r$.

We have recently developed methods which enable us to perform phase space calculations for certain bimolecular systems composed of symmetric polyatomic species, while rigorously conserving angular momentum.²⁰ It is the purpose of this paper to present exact classical expressions for phase space volumes for these systems. We place no restrictions on the magnitude of either the orbital, rotational, or total angular momentum, other than those imposed by energy and momentum conservation requirements. The formalism is developed to include bimolecular systems composed of a spherical top paired with a linear, spherical, or symmetric top molecule.

The general approach used here is straightforward. The system motion is separated into three parts—rotational, orbital, and vibrational. Next, the contribution to the system phase space due to rotational motion is evaluated. This result is then coupled to the system orbital motion to obtain the total nonvibrational phase space. We follow this approach because energy is a quadratic function of the molecular rotational angular momentum. For this reason the molecular rotational part of the problem is soluble in closed form. However, for the intermolecular potentials under consideration here, energy is a cubic (neutral–neutral) or quartic (ion–neutral) function of the system orbital angular momentum. Hence, this part of the problem is not completely soluble in closed form. The equations we obtain can easily be evaluated using numerical methods, however.

The organization of the paper is as follows. In Sec. II we define the phase space of interest and present a brief outline of our approach. In Sec. III we derive expressions for the sum and density of rotational states of

a polyatomic bimolecular system. In Sec. IV these expressions are used to develop formulas for the total system phase space and establish a correspondence between our formulas and other expressions for rotational state sums and densities. In Sec. V we present a discussion and the results of some preliminary calculations on a model system.

II. GENERAL THEORY

Consider now a system composed of two symmetric polyatomic molecules with total energy E above the zero point of the potential energy surface and total angular momentum $\mathcal{J}$. The energy E is assumed to be partitioned into the total vibrational and rotational energies of the system, $\mathcal{E}_v$ and $\mathcal{E}_r$, and the relative translational energy $\mathcal{E}_t$,

$$E = \mathcal{E}_v + \mathcal{E}_r + \mathcal{E}_t.$$

It is convenient to also define the translational–rotational energy sum $\mathcal{E}_{tr}$ as

$$\mathcal{E}_{tr} = \mathcal{E}_t + \mathcal{E}_r. \quad (2.1)$$

The total accessible phase space is separated into the contributions from system vibrational motion and system rotational–translational motion. The vibrational phase space is treated semiclassically by the density of vibrational states $\rho(\mathcal{E}_v)d\mathcal{E}_v$. The rotational–translational phase space, or momentum space, is the number of energy accessible combinations of the rotational angular momentum vectors $\mathbf{J}_1$ and $\mathbf{J}_2$ and orbital angular momentum vector $\mathbf{L}$ which can produce the total angular momentum $\mathcal{J}$.

Let $\Gamma(\mathcal{E}_{tr}, \mathcal{J})$ be the momentum space accessible to the system as a function of $\mathcal{J}$ and of $\mathcal{E}_{tr}$. The total system phase space is, then

$$\Phi(E, \mathcal{J}) = \int_{\mathcal{E}_t^*}^E \rho(E - \mathcal{E}_{tr}) \Gamma(\mathcal{E}_{tr}, \mathcal{J}) d\mathcal{E}_{tr}. \quad (2.2)$$

The lower limit $\mathcal{E}_{tr}^*$ is the minimum translational–rotational energy which can generate a $\mathbf{J}_1\mathbf{J}_2\mathbf{L}$ set capable of producing the resultant $\mathcal{J}$. Likewise, let $\mathcal{O}(\mathcal{E}_t, \mathcal{E}_r, \mathcal{J})d\mathcal{E}_t$ be the momentum space accessible to the system as a function of $\mathcal{J}$, $\mathcal{E}_r$, and $\mathcal{E}_t$. The total system phase space at constant $\mathcal{E}_t$ is

$$\Phi'(E, \mathcal{E}_t, \mathcal{J}) d\mathcal{E}_t = d\mathcal{E}_t \int_{\mathcal{E}_r^*}^{E-\mathcal{E}_t} \rho(E - \mathcal{E}_t - \mathcal{E}_r) \mathcal{O}(\mathcal{E}_t, \mathcal{E}_r, \mathcal{J}) d\mathcal{E}_r. \quad (2.3)$$

The lower limit $\mathcal{E}_r^*$ is the minimum rotational energy which can generate a $\mathbf{J}_1\mathbf{J}_2$ set of magnitude large enough to couple with the maximum available $\mathbf{L}$ to produce $\mathcal{J}$. Both $\mathcal{E}_{tr}^*$ and $\mathcal{E}_r^*$ will be determined later.

In order to evaluate $\Gamma(\mathcal{E}_{tr}, \mathcal{J})$ and $\mathcal{O}(\mathcal{E}_t, \mathcal{E}_r, \mathcal{J})d\mathcal{E}_r$ the restraints imposed by angular momentum and energy conservation must be considered. The angular momentum restraints are treated in the following manner. The total angular momentum is given by the vector addition equation

$$\mathcal{J} = \mathbf{J}_1 + \mathbf{J}_2 + \mathbf{L}.$$

This equation is decomposed by first coupling $\mathbf{J}_1$ and $\mathbf{J}_2$ to obtain a resultant total rotational angular momentum vector $\mathbf{J}_r$. The resultant $\mathbf{J}_r$ is then coupled with $\mathbf{L}$ to

produce $\mathcal{J}$. The vector boundary on the momentum space is therefore

$$\mathcal{J} = \mathbf{J}_r + \mathbf{L}, \quad (2.4a)$$

where

$$\mathbf{J}_r = \mathbf{J}_1 + \mathbf{J}_2. \quad (2.4b)$$

Also, for a nonlinear symmetric molecule the range of K , the internal projection of J on the molecular symmetry axis, is given by (2.4c):

$$-J \leq K \leq J. \quad (2.4c)$$

The relations between energy and angular momenta provide additional restrictions on the available phase space. For an interaction potential with an attractive term of the general form $-C/r^n$, the maximum orbital angular momentum capable of overcoming the centrifugal barrier is¹²

$$L^* = [\Lambda \mathcal{E}_t]^{(n-2)/2n}, \quad (2.5a)$$

where

$$\Lambda = [(2/(n-2))C^{2/(n-2)}]^{n\mu/\hbar^2}]^{n/n-2} \quad (2.5b)$$

and μ is the reduced mass. For a given $\mathcal{E}_t$ and $\mathcal{E}_r$ the energy-angular momentum relations are given by Eqs. (2.6).

$$0 \leq L \leq L^*, \quad (2.6a)$$

$$\mathcal{E}_r = B_1 J_1^2 + B_2 J_2^2 \pm \lambda_1 K_1^2 \pm \lambda_2 K_2^2. \quad (2.6b)$$

In Eq. (2.6b), B is the degenerate rotational constant and λ is the difference in rotational constants associated with molecular oblateness or prolateness.²¹ For a sphere $\lambda_s = 0$. For an oblate top $\lambda_o = B_o - C_o$ and a negative sign is used in Eq. (2.6b). For a prolate top $\lambda_p = A_p - B_p$ and a positive sign is used. A_p and C_o are the nondegenerate rotational constants of a prolate and oblate top, respectively.

The classical integral forms for $\Gamma(\mathcal{E}_t, \mathcal{J})$ and $\mathcal{O}(\mathcal{E}_r, \mathcal{E}_r) d\mathcal{E}_r$ can now be written

$$\Gamma(\mathcal{E}_t, \mathcal{J}) = \int_V dK_1 dK_2 dJ_1 dJ_2 dJ_r dL, \quad (2.7a)$$

$$\mathcal{O}(\mathcal{E}_r, \mathcal{E}_r, \mathcal{J}) d\mathcal{E}_r = \int_{\Delta V} dK_1 dK_2 dJ_1 dJ_2 dJ_r dL. \quad (2.7b)$$

The subscript V indicates that the integral defines a hypervolume in momentum space, while ΔV indicates that the integral defines a differential hypervolume on the hypersurface of constant rotational energy $\mathcal{E}_r$.²² If a molecule is linear the momentum space is reduced by one dimension since K is absent for a linear molecule.

In order to determine $\Gamma(\mathcal{E}_t, \mathcal{J})$ and $\mathcal{O}(\mathcal{E}_r, \mathcal{E}_r) d\mathcal{E}_r$ the accessible volume or differential volume in $K_1 K_2 J_1 J_2$ space for an LJ_r pair of arbitrary magnitude is first determined. This result is then integrated over the accessible area in the $L-J_r$ plane. Consider first $\Gamma(\mathcal{E}_t, \mathcal{J})$. For a given L , Eq. (2.5a) defines a lower limit $\mathcal{E}_t^*$ on the translational energy:

$$\mathcal{E}_t^* = L^{2n/(n-2)}/\Lambda. \quad (2.8)$$

Equations (2.1) and (2.8) in turn define an upper limit on $\mathcal{E}_t$:

$$\mathcal{E}_t^* = \mathcal{E}_{tr} - \mathcal{E}_t^* \quad (2.9a)$$

$$\geq B_1 J_1^2 + B_2 J_2^2 \pm \lambda_1 K_1^2 \pm \lambda_2 K_2^2. \quad (2.9b)$$

The boundaries imposed by Eqs. (2.4b) and (2.9b) define $\Gamma(\mathcal{E}_t^*, J_r)$, the available volume in $K_1 K_2 J_1 J_2$ space,

$$\Gamma(\mathcal{E}_t^*, J_r) = \int_V dK_1 dK_2 dJ_1 dJ_2. \quad (2.10)$$

In order to determine $\Gamma(\mathcal{E}_t, \mathcal{J})$ the expression for $\Gamma(\mathcal{E}_t^*, J_r)$ is integrated over the accessible area in the $L-J_r$ plane. The bounds on this integration are given by Eqs. (2.1), (2.4a), (2.6a), and other boundaries which will be determined.

The differential volume $\mathcal{O}(\mathcal{E}_r, \mathcal{E}_r, \mathcal{J}) d\mathcal{E}_r$ is evaluated by the same procedure. Equations (2.4b) and (2.6b) define the differential volume $\mathcal{O}(\mathcal{E}_r, J_r) d\mathcal{E}_r$ as

$$\mathcal{O}(\mathcal{E}_r, J_r) d\mathcal{E}_r = \int_{\Delta V} dK_1 dK_2 dJ_1 dJ_2, \quad (2.11)$$

while Eqs. (2.4a), (2.6a), and other boundaries define the accessible area in the $L-J_r$ plane used to determine $\mathcal{O}(\mathcal{E}_r, \mathcal{E}_r, \mathcal{J}) d\mathcal{E}_r$.

The quantities $\Gamma(\mathcal{E}_r, J_r)$ and $\mathcal{O}(\mathcal{E}_r, J_r)$ have simple physical interpretations. The momentum space volume $\Gamma(\mathcal{E}_r, J_r)$ is the sum of rotational states of a bimolecular system with total rotational angular momentum J_r and rotational energy less than or equal to $\mathcal{E}_r$. The quantity $\mathcal{O}(\mathcal{E}_r, J_r)$ is the density of rotational states for an analogous system with rotational energy between $\mathcal{E}_r$ and $\mathcal{E}_r + d\mathcal{E}_r$. It is simply the derivative of $\Gamma(\mathcal{E}_r, J_r)$ with respect to rotational energy.

$$\mathcal{O}(\mathcal{E}_r, J_r) = \partial \Gamma(\mathcal{E}_r, J_r) / \partial \mathcal{E}_r \quad (2.12)$$

A similar interpretation can be given to $\Gamma(\mathcal{E}_t, \mathcal{J})$ and $\mathcal{O}(\mathcal{E}_t, \mathcal{E}_r, \mathcal{J})$.

III. SUM AND DENSITY OF ROTATIONAL STATES

A. General considerations

We are now ready to determine $\Gamma(\mathcal{E}_t^*, J_r)$ and $\mathcal{O}(\mathcal{E}_r, J_r)$. The method used is to exploit the geometrical interpretation of Eq. (2.10). That is, we first determine the geometrical structure of the momentum space volumes generated by the energy and angular momentum conservation requirements given by Eqs. (2.4), (2.6b), and (2.9). Then, using straightforward integral calculus, expressions for these volumes are determined. The expressions obtained are for $\Gamma(\mathcal{E}_t^*, J_r)$. The density of states, $\mathcal{O}(\mathcal{E}_r, J_r)$ is obtained from Eq. (2.12). Since phase space is a multidimensional position-momentum space, in this sense a geometrical approach is natural.

The only other use of solid geometry in phase space theory that we know of is the work of Dagdigian *et al.*¹⁶ They used 3-dimensional figures as aids in developing an algorithm to determine the quantum mechanical phase space for a pair of linear molecules. However, the classical theory will be valid in most applications to polyatomic molecules, since the molecular rotational spacing is small for large molecules. Also, by using classical theory we can accomplish mathematically what a computer must do in the quantum mechanical theory.

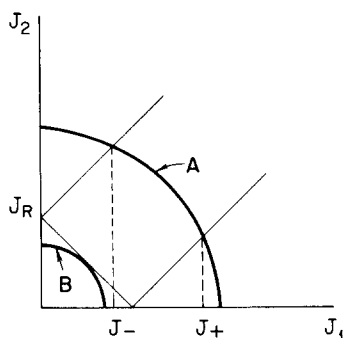


FIG. 1. Boundaries in the J_1J_2 plane. Curves A and B correspond to Eq. (2.4c). Curve B represents the minimum rotational energy for which there is available system phase space.

Before we actually determine the rotational state sums and densities some preliminary quantities must be derived. These are illustrated in Fig. 1, which is a J_1J_2 diagram. In this figure curves A and B are given by the energy boundary (2.6b). Since Fig. 1 is in the J_1J_2 plane, K_1 and K_2 are zero. The rectangular boundaries are the momentum limits imposed by (2.4b);

$$J_1 = |J_2 \pm J_r|.$$

The first quantity we are interested in is the intersections of (2.4b) and (2.6b), labeled in Fig. 1 as $J_{\pm}$. These intersections are obtained by substituting (2.4b) into (2.6b). After some simple rearranging we obtain

$$J_{\pm} = [(\omega \mathcal{E}_r - B_1 B_2 J_r^2)^{1/2} \pm B_2 J_r] / \omega,$$

where $\omega = B_1 + B_2$.

The other preliminary intersection needed is for $\mathcal{E}_m$, the minimum energy for a given J_r for which there is available momentum space. This situation is illustrated by curve B. To obtain $\mathcal{E}_m$, simply set J_+ equal to J_- to obtain

$$\mathcal{E}_m = B_r J_r^2,$$

where B_r is the system reduced rotational constant

$$B_r = B_1 B_2 / \omega.$$

B. The determination of $\Gamma(\mathcal{E}_r^*, J_r)$ and $\mathcal{P}(\mathcal{E}_r, J_r)$

It is appropriate to now determine $\Gamma(\mathcal{E}_r^*, J_r)$ for a pair of spherical top molecules. First, recall that $\mathcal{E}_r^*$ and J_r are constant and K_1, K_2, J_1 , and J_2 are the variables whose range must be determined. Consider first $K_2 = 0$. If $J_r > (\mathcal{E}_r^*/B_1)^{1/2}$ the momentum space volume of interest is given in Fig. 2(a). The J_1 and J_2 axes in this figure correspond to the equivalent labeled axes in Fig. 1. That is, Fig. 1 is just a two dimensional slice of Fig. 2(a) taken at $K_1 = 0$. [More correctly, if Figs. 1 and 2(a) were to coincide exactly, the energy boundary in Fig. 1 should approach curve A near $J_1 = 0$ and curve B near $J_2 = 0$.]

It is useful to use Fig. 1 to further interpret Fig. 2(a). The triangular planes in Fig. 2(a) are simply extensions of the rectangular momentum boundaries of Fig. 1 into the K_1 dimension. The curved surface in Fig. 2(a) is the energy boundary (2.6b). If $\mathcal{E}_r^*$ is increased in Fig. 2(a) this surface would soon become identical to curve A of Fig. 1. In this case the lower

triangle in Fig. 2(a) would extend to the K_1J_1 plane and then "travel upward" parallel to the upper triangular plane and at a distance $2J_r$ from it.

So far we have only considered $K_2 = 0$. If K_2 is increased, however, another dimension must be added to Fig. 2(a). Of course we cannot easily perceive a four dimensional space geometrically. However, if one variable is held constant the other three can be easily perceived. This is just what is done in Fig. 2(a) by setting $K_2 = 0$. In fact, this procedure is used implicitly to separate Eqs. (2.7a) and (2.7b) into (2.10) and (2.11).

Consider now a constant J_1 . The energy and momentum restrictions produce the $K_1K_2J_2$ -space boxlike structure shown in Fig. 2(b). This structure is the geometrical representation of the available system momentum space at constant $\mathcal{E}_r^*, J_r$, and J_1 . The end faces of this box correspond to $|K_1| = J_1$ and the sloped faces correspond to $|K_2| = J_2$. For a given J_1 the top and bottom faces in Fig. 2(b) correspond to the maximum and minimum values J_2 can have within the rectangle in Fig. 1. The minimum value of J_2 , that is the bottom face,

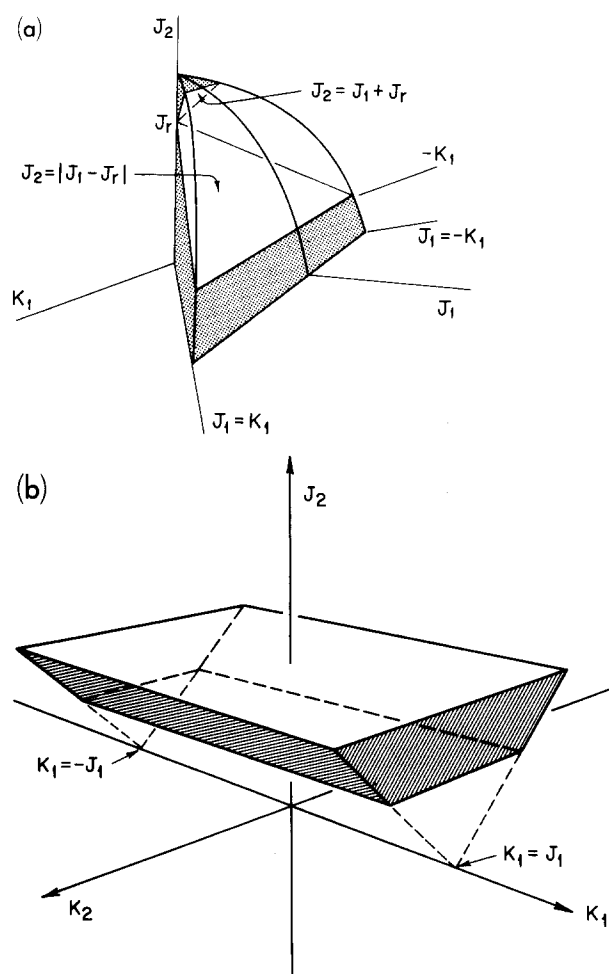


FIG. 2. (a) Available phase space in $K_1J_1J_2$ space for the sphere-sphere system when $[\mathcal{E}_r^*/B_2]^{1/2} > J_r > [\mathcal{E}_r^*/B_1]^{1/2}$. For the sphere-linear system $K_1J_1J_2 = K_2J_rJ_1$. (b) Available phase space in $K_1K_2J_2$ space. For $J_1 < J_-$ all boundaries are due to momentum restrictions. When $J_+ > J_1 > J_-$ the top boundary is determined from energy conservation. This is explained in the text.

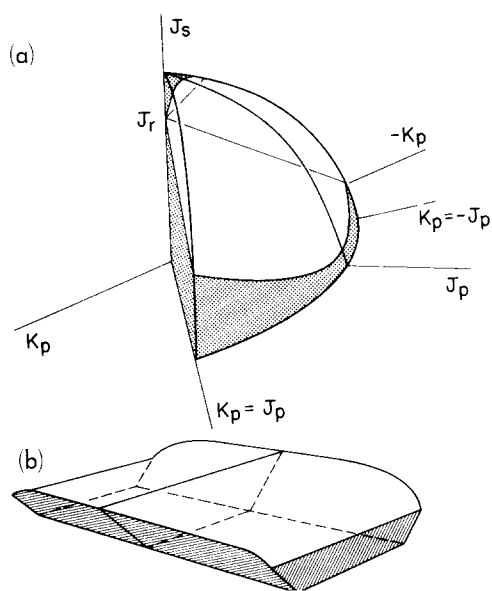


FIG. 3. (a) Available phase space in $K_p J_p J_s$ space for the sphere-prolate system when $[\mathcal{E}_r^*/B_s]^{1/2} > J_r > [\mathcal{E}_r^*/B_p]^{1/2}$. The curvature is due to the dependence of rotational energy on K_p . (b) An example of available phase space in $K_s J_s J_p$ space for the sphere-prolate system. This figure corresponds to Fig. 2(b) with the axes removed for clarity. The curvature is due to the energy dependence of K_p .

is $J_2 = |J_1 - J_r|$. For J_1 in the range $0 \leq J_1 \leq J_-$ the maximum value of J_2 , that is the top face, is $J_2 = J_1 + J_r$. For J_1 in the range $J_- \leq J_1 \leq J_+$ the maximum value of J_2 is $J_2 = (\mathcal{E}_r^* - B_1 J_1^2)^{1/2} / B_2^{1/2}$.

In order to determine $\Gamma(\mathcal{E}_r^*, J_r)$ we have to integrate the box in Fig. 2(b) over all available values of J_1 . In the region $J_1 \approx 0$ the range of values of J_2 is clustered around J_r . This can be seen from inspection of either Figs. 1 or 2(a). The maximum value of K_2 will also be approximately equal to J_r , while the maximum K_1 will be near 0. Therefore at $J_1 = 0$ the box begins as a line at $J_2 = J_r$, in the $K_2 J_2$ plane, parallel to the K_2 axis, and with length equal to $2J_r$. As J_1 increases, K_1 , K_2 , and J_2 all increase in range, that is, the box grows. When J_1 is less than J_r the box is wider at its midpoint along the K_2 direction than along the K_1 direction. If the values of $\mathcal{E}_r^*$ and J_r are such that J_1 can become larger than J_r , then when J_1 passes J_r the length of the midpoint in the K_2 direction becomes equal to the length in the K_1 direction. When J_1 reaches J_- the box begins to collapse along the J_2 axis. The collapse occurs because the maximum value of J_2 is now along curve A of Fig. 1. When $J_1 = J_+$, the top face of the box merges with the bottom face and the box disappears.

Now that we have seen the qualitative geometric development of $\Gamma(\mathcal{E}_r^*, J_r)$ for a pair of spheres, the problem must be put on an analytical basis. What must be done is to determine an expression for the volume of the "phase space" box as a function of J_1 and integrate this expression over the accessible range of J_1 . In order to determine the box volume consider the trapezoid formed by slicing the box in the $K_2 J_2$ plane at a constant K_1 . The lower bound of the trapezoid is J_2

$= |J_1 - J_r|$ and the upper bound may be either $J_2 = J_1 + J_r$ or the energy boundary $J_2 = (\mathcal{E}_r^* - B_1 J_1^2)^{1/2} / B_2^{1/2}$. The sides of the trapezoid have slope equal to $|1|$. If the upper bound is $J_2 = J_1 + J_r$ the area is

$$A_1 = 4J_r J_1. \quad (3.1a)$$

If the upper bound is the energy boundary the area becomes

$$A_2 = \mathcal{E}_r^* / B_2 - J_r^2 + 2J_r J_1 - \omega J_1^2 / B_2. \quad (3.1b)$$

For $0 \leq J_1 \leq J_-$ the box volume is

$$\begin{aligned} V_1(J_1) &= \int_0^{J_1} A_1 dK_1 \\ &= 8J_r J_1^2, \end{aligned} \quad (3.2a)$$

while for $J_- \leq J_1 \leq J_+$ the volume becomes

$$\begin{aligned} V_2(J_1) &= 2 \int_0^{J_1} A_2 dK_1 \\ &= 2[\mathcal{E}_r^* / B_2 - J_r^2] J_1 + 4J_r J_1^2 - 2\omega J_1^3 / B_2. \end{aligned} \quad (3.2b)$$

We can now integrate the box volumes in Eq. (3.2) over the range of J_1 . There are two cases to consider depending on whether $\mathcal{E}_r^*$ is greater or lesser than $B_r J_r^2$. For $\mathcal{E}_r^* \geq B_r J_r^2$ the integration over J_1 has the form

$$\Gamma(\mathcal{E}_r^*, J_r) = \int_0^{J_-} V_1(J_1) dJ_1 + \int_{J_-}^{J_+} V_2(J_1) dJ_1,$$

while for $B_r J_r^2 \geq \mathcal{E}_r^* \geq B_r J_r^2$ the integration becomes

$$\Gamma(\mathcal{E}_r^*, J_r) = \int_{J_-}^{J_+} V_2(J_1) dJ_1.$$

The result of both integrations, and therefore the result for the entire range $\mathcal{E}_r^* \geq B_r J_r^2$, is

$$\Gamma(\mathcal{E}_r^*, J_r) = (8J_r / 3\omega^3) (\omega \mathcal{E}_r^* - B_1 B_2 J_r^2)^{3/2}. \quad (3.3a)$$

Using the same geometric procedures it is possible to derive classical expressions for $\Gamma(\mathcal{E}_r^*, J_r)$ for a spherical top paired with either a linear molecule or a symmetric top. For the sphere-linear system the substitution $K_1 J_1 J_2 = K_s J_s J_l$ is made. Now K_2 is always zero, since a linear molecule has no internal projection. Therefore we need only consider three dimensional volumes such as the one shown in Fig. 2(a). For the sphere-prolate system, the substitution $K_1 K_2 J_1 J_2 = K_p K_s J_p J_s$ is made. The internal projection of the prolate top is a function of energy. This fact introduces curvature into the momentum space volumes. The results are shown in Fig. 3. In Fig. 3(b) the volume pictured is only one of five different forms this structure may have. For the sphere-oblate system, the curvature is in the opposite direction as shown in Fig. 4. Again, in Fig. 4(b) there are five possible structures. The derivation of $\Gamma(\mathcal{E}_r^*, J_r)$ for these three systems is given elsewhere.²³ Only the results are presented here:

sphere-linear,

$$\begin{aligned} \Gamma(\mathcal{E}_r^*, J_r) &= (2J_r / \omega^2) (\omega \mathcal{E}_r^* - B_s (B_l + B_s / 3) J_r^2), \quad \mathcal{E}_r^* \geq B_s J_r^2 \\ &= 4 / 3 B_s \omega^2 (\omega \mathcal{E}_r^* - B_s B_l J_r^2)^{3/2}, \quad B_s J_r^2 \geq \mathcal{E}_r^* \geq B_r J_r^2; \end{aligned} \quad (3.3b)$$

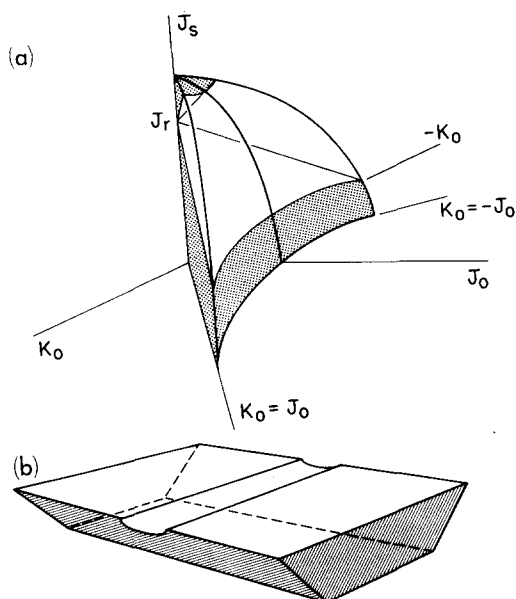


FIG. 4. (a) Available phase space in $K_r J_0 J_s$ space for the sphere-oblate system when $[\delta_r^*/B_s]^{1/2} > J_r > [\delta_r^*/C_0]^{1/2}$. The curvature is due to the dependence of rotational energy in K_0 . (b) An example of available phase space in $K_s J = J_0$ space for the sphere-oblate system.

sphere-prolate,

$$\Gamma(\delta_r^*, J_r) = G_1(\delta_r^*, J_r) + G_2(\delta_r^*, J_r) \theta_\Delta(\delta_r^*, J_r), \quad \delta_r^* \geq B_\eta J_r^2 \\ = (\pi/2) G_2(\delta_r^*, J_r), \quad B_\eta J_r^2 \geq \delta_r^* \geq B_s J_r^2, \quad (3.3c)$$

where

$$G_1(\delta_r^*, J_r) = [5\eta\omega\delta_r^* - (3B_p\eta + 2A_p\omega)B_s J_r^2] \\ \times (\eta\delta_r^* - B_s A_p J_r^2)^{1/2} J_r / 3\eta^2 \omega^2, \\ G_2(\delta_r^*, J_r) = (\omega\delta_r^* - B_s B_p J_r^2)^2 / \lambda_p^{1/2} \omega^{5/2} B_s, \\ \theta_\Delta(\delta_r^*, J_r) = \frac{1}{2}(\sin^{-1}\chi_+ - \sin^{-1}\chi_-), \\ = \sin^{-1}\chi_0 - \sin^{-1}\chi_-,$$

and

$$\chi_\pm = \frac{\omega(\eta\delta_r^* - B_s A_p J_r^2)^{1/2} \pm \lambda_p B_s J_r}{\eta(\omega\delta_r^* - B_s B_p J_r^2)^{1/2}}, \\ \chi_0 = (\omega/\eta)^{1/2},$$

and

$$\eta = B_s + A_p, \quad B_\omega = B_s B_p / \omega, \quad B_\eta = B_s A_p / \eta;$$

sphere-oblate,

$$\Gamma(\delta_r^*, J_r) = G_1(\delta_r^*, J_r) + G_2(\delta_r^*, J_r) \mathcal{L}(\delta_r^*, J_r), \quad \delta_r^* \geq B_\eta J_r^2. \quad (3.3d)$$

Here, $G_1(\delta_r^*, J_r)$ and $G_2(\delta_r^*, J_r)$ have the same form as in Eq. (3.3c) with B_p and A_p replaced by B_0 and C_0 , respectively. The function $\mathcal{L}(\delta_r^*, J_r)$ is defined as

$$\mathcal{L}(\delta_r^*, J_r) = \ln \frac{\omega^{1/2}(\eta\delta_r^* - B_s C_0 J_r^2)^{1/2} + \lambda_0^{1/2} B_s J_r}{\eta^{1/2}(\omega\delta_r^* - B_s B_0 J_r^2)^{1/2}}. \quad (3.3e)$$

In order to determine $\mathcal{P}(\delta_r^*, J_r)$ we simply apply Eq. (2.12) to $\Gamma(\delta_r^*, J_r)$. The results are

sphere-sphere,

$$\mathcal{P}(\delta_r^*, J_r) = (4J_r/\omega^2)(\omega\delta_r^* - B_1 B_2 J_r^2)^{1/2}, \quad (3.4a)$$

sphere-linear,

$$\mathcal{P}(\delta_r^*, J_r) = 2J_r/\omega, \quad B_s J_r^2 \geq \delta_r^* \geq B_s J_r^2 \\ = \frac{2}{B_s \omega} (\omega\delta_r^* - B_s B_1 J_r^2)^{1/2}, \quad \delta_r^* \leq B_s \leq B_s J_r^2; \quad (3.4b)$$

sphere-prolate,

$$\mathcal{P}(\delta_r^*, J_r) = \frac{2J_r}{\eta\omega} (\eta\delta_r^* - B_s A_p J_r^2)^{1/2} + \frac{2(\omega\delta_r^* - B_s B_p J_r^2)}{\lambda_p^{1/2} \omega^{3/2} B_s} \\ \times \theta_\Delta(\delta_r^*, J_r), \quad \delta_r^* \geq B_\eta J_r^2 \\ = \pi \frac{(\omega\delta_r^* - B_s B_p J_r^2)}{\lambda_p^{1/2} \omega^{3/2} B_s}, \quad B_\eta J_r^2 \geq \delta_r^* \geq B_\omega J_r^2; \quad (3.4c)$$

sphere-oblate,

$$\mathcal{P}(\delta_r^*, J_r) = \frac{2J_r}{\eta\omega} (\eta\delta_r^* - B_s C_0 J_r^2)^{1/2} + \frac{2(\omega\delta_r^* - B_s B_0 J_r^2)}{\lambda_0^{1/2} \omega^{3/2} B_s}, \\ \times \mathcal{L}(\delta_r^*, J_r), \quad \delta_r^* \geq B_\eta J_r^2. \quad (3.4d)$$

As far as we know, Eqs. (3.3) and (3.4) are the only available results for rotational state densities and sums for bimolecular systems with a given total rotational angular momentum. In the next section, where we consider $\mathbf{J}_r \mathbf{L}$ coupling, it will be shown that these formulas give the correct results when the restraints imposed by angular momentum conservation are removed. We will also establish the correspondence between our formulas and those presented by Klotz for unimolecular reaction rate theory.

IV. INTEGRATION IN THE L - J_r PLANE

A. Integration of $\Gamma(\delta_r^*, J_r)$

In order to integrate $\Gamma(\delta_r^*, J_r)$ in the L - J_r plane it is necessary to determine the boundaries on L and J_r as a function of g and δ_{tr} . The momentum boundaries are quite straightforward, and are given by Eq. (2.4a). The limits on L and J_r as a function of δ_{tr} are obtained by combining Eqs. (2.1), (2.5), and $\delta_m = B_r J_r^2$,

$$L^{2n/n-2} / \Lambda + B_r J_r^2 \leq \delta_{tr}. \quad (4.1)$$

For the sphere-oblate and sphere-prolate systems B_r is given by B_η and B_ω , respectively. For the sphere-linear and sphere-prolate systems the integration is separated into two regions due to the two functional forms of $\Gamma(\delta_r^*, J_r)$. This division is illustrated in Fig. 5(a).

The value of δ_{tr}^* , the minimum translational rotational energy capable of generating an L - J_r pair capable of producing g , is obtained by setting Eq. (4.1) tangent to the line $g = L + J_r$. When the eliminated variable is J_r this procedure gives

$$\frac{n}{n-2} \frac{1}{\Lambda} L^{\dagger n+2/n-2} - B_r (g - L^\dagger) = 0, \quad (4.2)$$

where $L^\dagger$ is the value of L at $\delta_{tr} = \delta_{tr}^*$. There are two cases of interest; neutral-neutral reactions and ion-neutral reactions. For neutral-neutral reactions, $n=6$ and the solution to Eq. (4.2) is

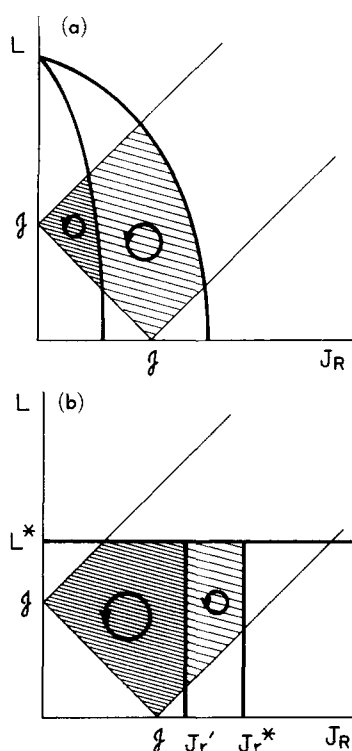


FIG. 5. (a) Range of integration in the L - J_r plane for $\Gamma(\epsilon_r, J_r)$. The curves are the boundaries given by Eq. (4.1). The circles correspond to the path of integration of $I(J_r)$. This figure is for the sphere-prolate and sphere-linear systems. For the sphere-sphere and sphere-oblate systems only one boundary exists. (b) Range of integration in the L - J_r plane for $\phi(E_r, E_r, J)$. The circles correspond to the integration path for $I(J_r)$. This figure is for the sphere-prolate and sphere-linear systems. For the sphere-sphere and sphere-oblate systems only one boundary exists.

$$L_{\text{neutral}}^{\dagger} = \frac{B_r \Lambda}{3} \left[\left(1 + \frac{6J}{B_r \Lambda} \right)^{1/2} - 1 \right]. \quad (4.3)$$

For ion-neutral reactions $n=4$ and Eq. (4.2) has the solution

$$L_{\text{ion}}^{\dagger} = (B_r \Lambda / 2)^{1/3} \{ (S + J/2)^{1/3} - (S - J/2)^{1/3} \}, \quad (4.4)$$

where

$$S = (J^2/4 + B_r \Lambda / 54)^{1/2}.$$

The energy minimum $\epsilon_{\text{tr}}^{\dagger}$ is given by

$$L^{2n/(n-2)}/\Lambda + B_r (J - L^{\dagger})^2 = \epsilon_{\text{tr}}^{\dagger}. \quad (4.5)$$

For the sphere-linear and sphere-prolate systems there is a second energy minimum ϵ'_{tr} given by Eqs. (4.2)–(4.5) with B_r replaced by B_s or B_n .

In order to evaluate $\Gamma(\epsilon_{\text{tr}}, J)$ the integration over J_r is done first, since for all systems presented here the integration of $\Gamma(\epsilon_r^*, J_r)$ can always be performed in closed form. The results for the sphere-oblate and sphere-prolate systems are presented elsewhere.²³ Numerical techniques must be used to obtain the intersections of Eqs. (2.4a) and (4.1) and to integrate $\Gamma(\epsilon_r^*, J_r)$ over L .

The integration can be viewed as a simple application of Green's theorem to the integration of a single function in a closed region bounded by a contour C . Define the result of the integration over J_r as

$$I(J_r) \equiv \int \Gamma(\epsilon_r^*, J_r) dJ_r,$$

or

$$\Gamma(\epsilon_r^*, J_r) = \partial I(J_r) / \partial J_r, \quad (4.6)$$

and apply Green's theorem,

$$\begin{aligned} \int_A \int \Gamma(\epsilon_r^*, J_r) dJ_r dL &= \int_A \int \frac{\partial I(J_r)}{\partial J_r} dJ_r dL \\ &= \int_C I(J_r) dL. \end{aligned} \quad (4.7)$$

The contour consists of the energy and momentum restrictions given by Eqs. (2.4a) and (4.1). As an example consider the integration of $\Gamma(\epsilon_r^*, J_r)$ for the sphere-sphere and sphere-oblate systems when $\epsilon_{\text{tr}} \geq B_r J^2$ and $\epsilon_{\text{tr}} \geq J^{2n/(n-2)}/\Lambda$. We apply Green's theorem to obtain

$$\Gamma(\epsilon_{\text{tr}}, J) = \int_0^{L_-} I(J + L) dL + \int_{L_+}^0 I(|J - L|) dL,$$

where $L_{\pm}$ are the intersections of Eqs. (2.4a) and (4.1). In the evaluation of the contour integral we have made use of the fact that for the sphere-sphere and sphere-oblate systems $I[(\epsilon_r^*/B_r)^{1/2}] = 0$. There are two cases total for the sphere-sphere and sphere-oblate systems and five cases total for the sphere-prolate and sphere-linear systems. The entire set of integrations for $\Gamma(\epsilon_{\text{tr}}, J)$ for all four systems are presented elsewhere.²³

B. Integration of $\phi(\epsilon_r, J_r)$

The restrictions on the integration of $\phi(\epsilon_r, J_r)$ in the L - J_r plane are given by Eqs. (2.4a), (2.5), and (3.3). For the sphere-linear and sphere-prolate systems there is again a boundary in the integration region at $B_s J_r^2$ and $B_n J_r^2$. Typical examples of the region of integration are given in Fig. 5(b). The quantities J_r^* and J_r' are defined by

$$J_r^* = (\epsilon_r/B_r)^{1/2}, \quad (4.8)$$

$$J_r' = (\epsilon_r/B_s)^{1/2}, \quad \text{or} \quad (\epsilon_r/B_n)^{1/2}.$$

The value of $\epsilon_r^{\dagger}$ the minimum rotational energy needed to produce a J_r large enough to couple with L^* and give J , can now be determined. For $L^* \geq J$ all J_r^* can couple with L^* to give J . Therefore in this case $\epsilon_r^{\dagger}$ is

$$\epsilon_r^{\dagger} = 0, \quad L^* \geq J. \quad (4.9a)$$

If $J \geq L^*$ the minimum J_r^* is given by the intersection of the boundary $L + J_r = J$ with $L = L^*$. $\epsilon_r^{\dagger}$ is now

$$\epsilon_r^{\dagger} = B_r (J - L^*)^2, \quad J \geq L^*. \quad (4.9b)$$

For the integration in the L - J_r plane there are five cases to consider for the sphere-sphere and sphere-oblate systems and 17 cases for the sphere-linear and sphere-prolate systems. These cases arise due to the different relative magnitudes of J , L^* , J_r^* , and J_r' . A number of examples are presented here. The entire set of integrations for the sphere-sphere and sphere-linear systems are given elsewhere.²³

First consider the sphere-sphere system in the region $J \geq J_r^*$ and $L^* \geq J + J_r^*$. We again use the definition

$$\phi(\epsilon_r, J_r) = \partial I(J_r) / \partial J_r,$$

and apply Green's theorem to obtain

$$\begin{aligned} \phi(\epsilon_r, J_r) &= \int_{J_r^*}^{J+J_r^*} I(J_r^*) dL + \int_{J+J_r^*}^{J-J_r^*} I(|J-L|) dL \\ &= \pi \epsilon_r^2 / 2(B_1 B_2)^{3/2}. \end{aligned} \quad (4.10)$$

Consider now the sphere-linear system in the region $J \geq J_r^*$ and $L^* \geq J + J_r^*$. We define $I_1(J_r)$ for $0 \leq J_r \leq J_r^*$ and $I_2(J_r)$ for $J_r' \leq J_r \leq J_r^*$. The result of the integration is

$$\begin{aligned} \mathcal{P}(\mathcal{E}_r, \mathcal{E}_r, J) = & \int_{J_r^*}^{J+J_r^*} I_2(J_r^*) dL + \int_{J+J_r^*}^{J+J_r'} I_2(L-J) dL + \int_{J+J_r'}^{J-J_r'} I_2(J_r') dL + \int_{J-J_r'}^{J-J_r^*} I_2(J-L) dL + \int_{J-J_r^*}^{J+J_r'} I_1(J_r') dL \\ & + \int_{J+J_r'}^{J-J_r^*} I_1(|J-L|) dL = \frac{4\mathcal{E}_r^{3/2}}{3B_s^{3/2}B_l}. \end{aligned} \quad (4.11)$$

C. Limiting forms of $\mathcal{P}(\mathcal{E}_r, \mathcal{E}_r, J)$ and $\Gamma(\mathcal{E}_r, J)$

Physically the two results (4.10) and (4.11) give the total rotational density of states of the respective systems under the condition that the only restrictions on the available phase space are energy restrictions. Therefore these results should be able to be derived by partition function inversion since this technique readily addresses itself to this type of restriction. For a linear molecule the classical partition function is

$$\begin{aligned} Q_l = & \int_0^\infty 2J e^{-B_l J^2/kT} dJ \\ = & kT/B_l. \end{aligned} \quad (4.12)$$

For a spherical top molecule the classical partition function is

$$\begin{aligned} Q_s = & \int_0^\infty (2J)^2 e^{-B_s J^2/kT} dJ \\ = & \sqrt{\pi} (kT)^{3/2}/B_s^{3/2}. \end{aligned} \quad (4.13)$$

Using standard Laplace transform techniques the sphere-sphere rotational density of states is

$$\begin{aligned} \rho(\mathcal{E}_r) = & Q_1 Q_2 \mathcal{E}_r^2 / [(kT)^3 \Gamma(3)] \\ = & \pi \mathcal{E}_r^2 / 2(B_1 B_2)^{3/2}, \end{aligned}$$

and the sphere-linear density of states is

$$\begin{aligned} \rho(\mathcal{E}_r) = & Q_s Q_l \mathcal{E}_r^{3/2} / [(kT)^{3/2} \Gamma(\frac{5}{2})] \\ = & 4 \mathcal{E}_r^{3/2} / 3 B_s^{3/2} B_l, \end{aligned}$$

where $\Gamma(a)$ is the gamma function of argument a . Hence, our results give the correct classical limit when the angular momentum restraints are removed.

It is also possible to obtain a correspondence between our formulas and the results of Klots.¹⁷ He derived approximate expressions for the phase space for bimolecular systems under the restriction that the total angular momentum J was small. This restriction is quite suitable for unimolecular reactions of ions with thermal rotational distributions. Now if $J \approx 0$, then $L \approx J_r$. In this case it is possible to integrate $\Gamma(\mathcal{E}_r^*, J_r)$ over J_r in the range $0 \leq J_r \leq (\mathcal{E}_r^*/B_r)^{1/2}$. For the sphere-sphere system the results are

$$\begin{aligned} \Gamma(\mathcal{E}_r, J) \approx & \frac{8}{3\omega^3} \int_0^{(\mathcal{E}_r^*/B_r)^{1/2}} J_r (\omega \mathcal{E}_r^* - B_1 B_2 J_r^2)^{3/2} dJ_r \\ = & 8 \mathcal{E}_r^{5/2} / 15 B_1 B_2 \omega^{1/2} \end{aligned} \quad (4.14)$$

which is identical to the result obtained by Klots¹⁷; the small contribution to $\mathcal{E}_r$ given by $\mathcal{E}_r^* = \mathcal{E}_r - J^{2n/n-2}/\Lambda$ is neglected. For the sphere-linear system

$$\Gamma(\mathcal{E}_r, J) \approx \frac{\mathcal{E}_r^{5/2} \sin^{-1}\left(\frac{B_s}{\omega}\right)^{1/2}}{2 B_s (B_s B_l)^{1/2}},$$

which is also identical to the result obtained by Klots for this system.

Using this procedure one can derive a Klots-type expression for the rotational state sum for the sphere-prolate top system under the assumption that $J \approx 0$. Simply integrate the appropriate expression for $\Gamma(\mathcal{E}_r^*, J_r)$, Eq. (3.3c), over all available J_r as was done in (4.14). The integration of $\Gamma(\mathcal{E}_r^*, J_r)$ for the sphere-prolate system is readily performed^{23,24} yielding

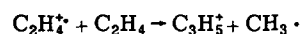
$$\Gamma(\mathcal{E}_r, J) \approx \frac{8}{15} \frac{\mathcal{E}_r^{5/2}}{B_s (B_s B_p \lambda_p)^{1/2}} \sin^{-1}\left(\frac{B_s \lambda_p}{A_p \omega}\right)^{1/2}. \quad (4.15)$$

Note that in the limit $\lambda_p \rightarrow 0$, that is as the prolate top approaches a spherical top, equation (4.15) reduces to (4.14).

V. DISCUSSION

In the previous sections expressions have been presented for both the sum of states and density of states as a function of energy for all pairwise combinations of a spherical top molecule paired with a linear molecule or with a spherical, oblate or prolate top molecule. These quantities will be applied to bimolecular²⁵ and unimolecular²⁶ ion-molecule reactions in subsequent papers. In the case of bimolecular reactions, expressions for both the energy dependence of the reaction cross section and of the product kinetic energy will be developed and these quantities calculated, and compared with experiment. In the case of unimolecular reactions, expressions for the energy dependence of the rate constant and the product kinetic energy will be developed, and these quantities calculated and compared with experiment. At this point, however, it is useful to determine the effect of approximating prolate or oblate top molecules as spherical top molecules. In practice it will often be necessary to do this. In addition, many systems will include asymmetric top molecules. At all times these molecules will have to be assumed to be either oblate, prolate, or spherical tops. Hence, we would like to have a feeling for the nature and magnitude of the errors these assumptions will introduce.

It is convenient and useful to use an actual ion-molecule reaction for our model calculation. We've chosen the reaction of ethylene ion with ethylene because we will be comparing calculations on this system



with experiment in later papers and because the system is complex enough yet well enough defined to be a useful test.

TABLE I. Rotational constants and neutral polarizabilities for species in the reaction $C_2H_4^+ + C_2H_4 \rightarrow C_3H_5^+ + CH_3^+$.

Species	A^a	B^a	C^a	α^b
C_2H_4	4.83	1.00	0.83	4.22
$C_2H_4^+$	4.83	1.00	0.83	
$C_3H_5^+$	1.53	0.30	0.48	
CH_3^+	9.63	9.63	4.82	2.2

^aRotational constants are in cm^{-1} and calculated as discussed in the text.

^bPolarizabilities are in units of $1 \times 10^{-24} cm^3$. The polarizabilities are calculated from the data of A. T. Amos and R. J. Crispin, J. Chem. Phys. 63, 1890 (1975).

The data used in the calculations are summarized in Table I. The rotational constants and polarizability of C_2H_4 are literature values. It is assumed $C_2H_4^+$ has the same structure as C_2H_4 . Both C_2H_4 and $C_2H_4^+$ are near prolate tops. The rotational constants of $C_3H_5^+$ and CH_3^+ were calculated from assumed structures. The $C_3H_5^+$ ion was assumed to be the allyl cation, $[CH_2CHCH_2]^+$ and normal bond lengths and reasonable bond angles were chosen consistent with this structure. The resulting rotational constants indicate $C_3H_5^+$ is a near prolate top. The CH_3^+ radical was assumed to be planar with D_{3h} symmetry, that is an oblate top. The polarizability of CH_3^+ was estimated from atomic polarizabilities.

In the following calculations it is necessary to approximate nonspherical molecules as spherical tops. This is accomplished by taking the geometric mean of the three rotational constants of a molecule. When an asymmetric molecule is approximated as a prolate top the geometric mean of the two rotational constants B and C is taken as the degenerate rotational constant.

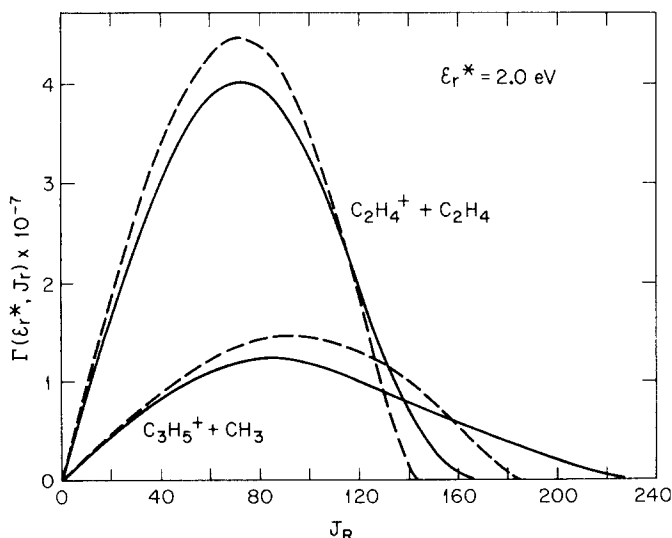


FIG. 6. An example of the error introduced into $\Gamma(\epsilon_r^*, J_r)$ by approximating a prolate spheroid as spherical. In the $C_3H_5^+$, CH_3^+ system, CH_3^+ is presumed spherical and $C_3H_5^+$ prolate. The dashed line represents the sphere-sphere calculation.

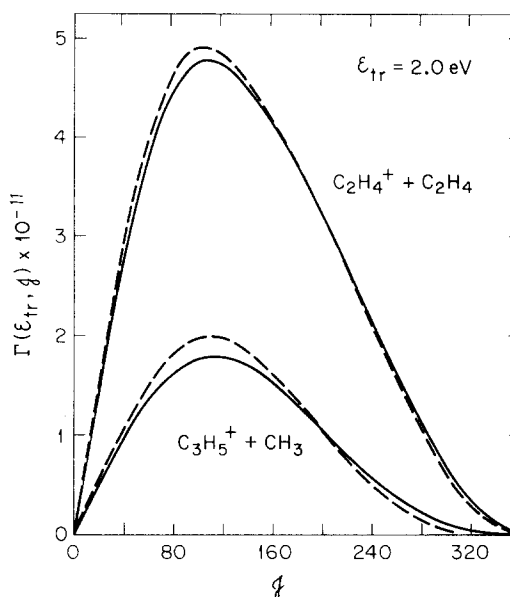


FIG. 7. An example of the error introduced into $\Gamma(\epsilon_{tr}, J)$ by approximating a prolate spheroid as spherical. The sphere-sphere curve in the $C_3H_5^+$, CH_3^+ system also corresponds to the sphere. The dashed line represents the sphere-sphere calculation.

Two types of calculations are carried out. The sphere-sphere approximation is compared with the sphere-prolate approximation in both cases. In Fig. 6 the sum of states, $\Gamma(\epsilon_r^*, J_r)$ is given as a function of J_r for a rotational energy limit $\epsilon_r^* = 2.0$ eV, for both the product and reactant pairs. The results clearly show that the approximation of a (near) prolate molecule as a sphere will tend to overestimate $\Gamma(\epsilon_r^*, J_r)$ for low J_r and underestimate it for high J_r . These effects compensate for each other in the total area under the curves, which remains fairly constant. The truth of this statement is apparent in the calculations in Fig. 7, where the integrated sum of states, $\Gamma(\epsilon_{tr}, J)$, is plotted as a function of total angular momentum J for an energy $\epsilon_{tr} = 2.0$ eV. This is an important point because $\Gamma(\epsilon_{tr}, J)$ is the quantity that will actually be used in cross section calculations.

Calculations were also performed treating $C_3H_5^+$ as a spherical top and CH_3^+ as an oblate top. The results were within 1% of the sphere-sphere results for $\Gamma(\epsilon_r^*, J_r)$ and were within 0.1% of the sphere-sphere results for $\Gamma(\epsilon_{tr}, J)$. This is not unexpected, since there is a limit to molecular oblateness, that is, the range of the rotational constant C is $B \geq C \geq B/2$, which imposes limits on the curvature shown in Fig. 4.

To summarize, using a geometrical approach in this paper we have derived formulas for the sum and density of rotational and rotational-orbital states for a pair of polyatomic symmetric molecules. With the results presented here it is now possible to perform accurate phase space calculations on reactions involving polyatomic species. Such calculations, in conjunction with accurate experimental studies, should prove to be quite valuable in diagnosing the mechanistic details of a chemical reaction involving polyatomic species. The

methods and formulas developed here should also prove valuable in determining the effect of molecular structure on any chemical process that is believed to proceed statistically. As an example, we have demonstrated the effect of molecular oblateness and prolateness on the phase space available to a given bimolecular system.

Finally, it should be mentioned that the sum and density of rotational states for other bimolecular systems, specifically two prolate spheroids and a prolate plus linear molecule, may also be determined using the geometrical methods presented here. However, the mathematics becomes more complex. The presence of a spherical molecule with a $2J_s$ degeneracy factor leads to areas in the $K_s J_s$ plane which are simple to integrate. It is this fact which produces relatively simple integrations over K and J of the other molecular species. The absence of this symmetry leads to multiple integrations of expressions involving square roots. Although tedious, these integrals may possibly be performed using techniques such as those presented elsewhere,²³ where $\Gamma(\delta_r^*, J_r)$ for the sphere-prolate system is integrated with respect to J_r .

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¹An excellent recent discussion is given in Faraday Discuss. Chem. Soc. 55. Also, Y. T. Lee, Talk given at the 23rd Annual Conference on Mass Spectrometry and Allied Topics, Houston, Texas, May (1975).

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